

R4: Uncertainty in the Known Draught γ

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How accurately must the floe draught be known?

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SHARC buoy project Code: `runR4GammaTolerance.m`, commit 0389536 Results: `Results/R4/R4_20261008_190753`

Rerun after the amplitude amendment (8 Oct). The EMM FRFs used by the 1 Oct run were low by $\alpha(\omega) = H\omega^2/g$; `computeSensorFRF` now multiplies by α and the twin cache was migrated exactly (`cfg.meta.changes`, 2026-10-08). Settings unchanged. This summary supersedes R4_20261001.183822. The physical (1% stiffness) tolerance barely moved; the significance tolerances tightened about fivefold because σ fell about sevenfold.

Summary

Question. The frozen inverse estimates $[\ln \beta, \ln R]$ and fixes the draught parameter γ at an assumed value. If that value is wrong by ϵ_γ , the fit absorbs the error through β and R . How large can ϵ_γ be? **Outcome.** A wrong γ leaks almost entirely into stiffness: per unit error in $\ln \gamma$, $\ln \hat{\beta}$ moves by 0.076 and $\ln \hat{R}$ by 0.007 (LSM6DSV16X). Keeping the stiffness error below 1% needs only $|\epsilon_\gamma| \leq 12.4\%$ (± 0.34 mm of draught at p_0). Keeping the bias below half a Fisher uncertainty unit ($d_{\text{sys}} \leq 0.5$) needs $|\epsilon_\gamma| \leq 0.31\%$ (± 0.009 mm), because the inverse is now so precise ($\sigma_{\ln \beta} = 0.050\%$) that a stiffness shift of 0.02% is already significant. **Practical consequence.** γ must come from *weighing* (mass and area), with the instrument mass included; even then the significance tolerance is at the limit of what weighing achieves, so draught error is a realistic systematic floor on the stiffness estimate, of order 0.05%, far inside the 1% accuracy target.

1 Question and setting

The draught parameter is $\gamma = d/H = \rho_i h / (\rho_w H)$. The inverse uses an assumed value $\gamma_{\text{inv}} = \gamma_{\text{true}}(1 + \epsilon_\gamma)$, so a relative error in γ is the same relative error in the draught d (or in the thickness h , with ρ_i known). Evaluated at p_0 with the E1 production setting: Level 2C with the reference at $0.9R$, $N = 65536$ at $\Delta t = 0.1$ s, Hann $L = 2048$, 84 bins, JONSWAP $H_s = 0.05$ m, $\omega_p = 5$ rad/s. All six pre-registered noise cases; headline LSM6DSV16X. At p_0 the draught is $\gamma_0 H = 2.74$ mm.

2 Method

1. Sensitivity to γ . E1 computes $J_\gamma = \partial f / \partial \ln \gamma$ by central differences as a *nuisance* sensitivity; it never enters the 2×2 Fisher matrix F .

2. Linear projection. The feature bias

$$b_\gamma = f(\gamma_{\text{true}}) - f(\gamma_{\text{inv}}) \approx -J_\gamma \ln(1 + \epsilon_\gamma)$$

is mapped to $\delta \theta_\gamma = F^{-1} J^T \Sigma^{-1} b_\gamma$ and $d_{\text{sys}} = \sqrt{\delta \theta_\gamma^T F \delta \theta_\gamma}$, each noise case with its own Σ_k , F and accepted bins. Pre-registered grid $\epsilon_\gamma = \pm 0.5, \pm 1, \pm 2, \pm 5, \pm 10, \pm 20\%$.

3. Tolerances. On a fine grid ($\pm 30\%$, 1201 points), the smallest $|\epsilon_\gamma|$ over both signs at which each criterion is first reached: $|\delta \beta / \beta| = 1\%$ (accuracy, $\delta \beta / \beta = e^{\delta \ln \beta} - 1$); $d_{\text{sys}} = 0.5$ and $d_{\text{sys}} = 1$ (significance).

4. Nonlinear model check (headline noise, $\epsilon_\gamma = \pm 5\%$ and $\pm 20\%$). The EMM is solved exactly at γ_{inv} and the wrong- γ model is fitted to the noiseless true features by fixed-Jacobian Gauss-Newton. Runtime 342 s (eight new EMM fields for the check; the rest from the cache).

3 Results

Tolerances by noise case.

Noise case	$\sigma_{\ln \beta}$	d_W	per unit $\ln \gamma$ error		tolerance $ \epsilon_\gamma $			$d_{\text{sys}} \leq 0.5$
			$\delta \ln \beta$	$\delta \ln R$	$ \delta \beta / \beta \leq 1\%$	$d_{\text{sys}} \leq 0.5$	$d_{\text{sys}} \leq 1$	draught (mm)
Datasheet grade	0.020%	1.209	0.0784	0.00657	12.03%	0.12%	0.25%	± 0.003
Three IMUs fused	0.043%	0.487	0.0756	0.00672	12.45%	0.27%	0.54%	± 0.007
LSM6DSV16X	0.050%	0.436	0.0757	0.00687	12.43%	0.31%	0.63%	± 0.009
G3 assumption	0.061%	0.391	0.0764	0.00709	12.33%	0.38%	0.76%	± 0.010
ICM-42688-P	0.063%	0.387	0.0765	0.00712	12.31%	0.39%	0.78%	± 0.011
LSM6DS3TR-C	0.150%	0.237	0.0806	0.00787	11.72%	0.86%	1.72%	± 0.024

Rows ordered from quietest to noisiest. The 1% stiffness tolerance corresponds to ± 0.32 to 0.34 mm of draught for every case. Draught values at p_0 (2.74 mm). Pre-amendment (1 Oct), LSM6DSV16X: sensitivity 0.0896, tolerances 10.61%, 1.64%, 3.26%.

Nonlinear model check (LSM6DSV16X).

ϵ_γ	$\delta\beta/\beta$		nonlinear / linear -1		difference [$\ln \beta, \ln R$] in σ	exact vs linear feature error
	linear	nonlinear	$\ln \beta$	$\ln R$		
-20%	-1.675%	-1.503%	-10.3%	-11.4%	[+3.50, +2.10]	10.5%
-5%	-0.388%	-0.378%	-2.5%	-2.8%	[+0.20, +0.12]	2.6%
+5%	+0.370%	+0.379%	+2.5%	+2.8%	[+0.18, +0.11]	2.5%
+20%	+1.390%	+1.525%	+9.7%	+11.0%	[+2.67, +1.65]	9.9%

Each case converged in 3 iterations. The relative error grows about linearly with $|\epsilon_\gamma|$ (about 0.5% per 1% of γ error), as before; the same relative error is now a larger fraction of the much smaller σ .

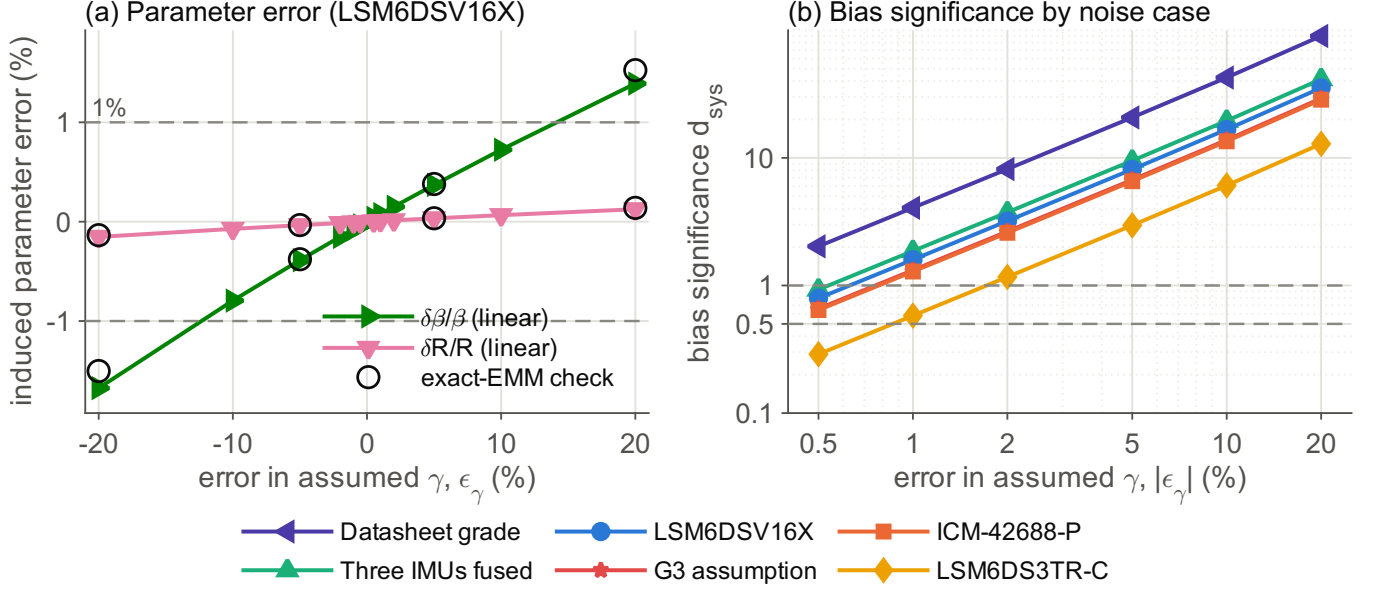


Figure 1: R4 sensitivity of the recovered parameters to the error in the known draught parameter γ (corrected FRF). (a) Induced relative errors in β and R for the LSM6DSV16X, with exact-EMM check points as open circles and $\pm 1\%$ stiffness thresholds dashed. (b) Bias significance d_{sys} against the magnitude of the γ error for every noise case; dashed lines at $d_{\text{sys}} = 0.5$ and 1 .

4 Interpretation

- A wrong draught biases stiffness, not radius.** $\delta \ln \beta / \delta \ln R \approx 11$: a 10% error in γ moves β by about 0.8% and R by about 0.07%. γ sets the plate's inertia, which the fit compensates mainly through its stiffness. The amendment reweighted the band towards high frequencies and lowered the stiffness sensitivity from 0.090 to 0.076.
- The two criteria have separated.** The 1% criterion depends only on the model geometry and is almost the same for every noise case (11.7 to 12.5%). The d_{sys} criterion is measured against σ , which is now 0.02 to 0.15%, so it demands 0.12 to 0.86%. A 0.31% draught error moves β by only 0.024%, physically negligible but half an uncertainty unit. With this much precision, the significance criterion asks for a calibration better than the physical problem needs.
- The tolerances are trustworthy.** The d_{sys} tolerances (0.12 to 0.86%) lie far inside $\pm 5\%$, where the exact EMM differs from the linear projection by 2.5% relative; at $|\epsilon_\gamma| = 0.3\%$ that gap is a negligible fraction of σ . The 1% stiffness tolerance (12.4%) is reached on the negative side, where the exact shift is about 6% smaller than the linear one, so the linear value is slightly conservative (exact about 13%).
- The bias budget is shared.** The tolerances above use the draught bias alone. With $d_W = 0.436$ already spent at the production setting, the Identifiable class ($d_{\text{tot}} \leq 0.5$, biases added as vectors) leaves even less room for γ error, depending on the relative direction of the two biases (per- ϵ classes in `R4_sweep.csv`). This is the same mechanism as D2 and R1: precision exposes every fixed bias.
- How to determine γ .** ± 0.009 mm of draught cannot be measured directly, but

$$\gamma = \frac{m}{\rho_w \pi R_{\text{phys}}^2 H}, \quad \left(\frac{\delta \gamma}{\gamma} \right)^2 \approx \left(\frac{\delta m}{m} \right)^2 + \left(2 \frac{\delta R_{\text{phys}}}{R_{\text{phys}}} \right)^2 + \left(\frac{\delta \rho_w}{\rho_w} \right)^2.$$

Weighing the 4.4 kg plate to ± 5 g (0.1%), measuring the radius to ± 0.5 mm (0.14% in $2\delta R/R$) and knowing the water density to 0.1% gives about 0.2%, inside the 0.31% tolerance. The earlier recipe (0.5% mass, ± 2 mm radius, about 0.8%) no longer meets the significance criterion, although it still gives a stiffness error of only about 0.06%.

6. **Instrument mass is now a required term.** The inverse assumes uniform areal mass. Four sensor packages of about 50 g each (not yet weighed) would add about 4.5% to the 4.4 kg plate, fifteen times the 0.31% tolerance and a third of the 1% stiffness tolerance. The packages must be weighed and included in γ , and their concentration at four points is itself a model error to be checked.

5 Limitations

- **One floe.** Evaluated at p_0 only; other floes have other sensitivities (A1).
- **d_{sys} alone.** See item 4: the combined class uses d_{tot} with the Welch bias.
- **Fixed layout and sea state.** Level 2C and the reference JONSWAP spectrum only.
- **Comparison with the earlier estimate.** An earlier single-FRF magnitude estimate gave 0.057 for β and 0.008 for R . The radius coupling agrees; the stiffness coupling is 1.3 times larger in the full transmissibility and Fisher treatment. Different observables, so a consistency check only.

Requirement for the experiment and the thesis

For the LSM6DSV16X at p_0 : γ within $\pm 12.4\%$ keeps the stiffness error below 1%; γ within $\pm 0.31\%$ keeps the draught bias below half an uncertainty unit. Determine γ by weighing (mass to about 0.1%, radius to about ± 0.5 mm), include the instrument mass, and report γ with its uncertainty alongside every recovered β . At this precision, draught calibration rather than sensor noise sets the systematic floor of the stiffness estimate.

Output	Contents
R4_20261008.190753.log / .mat / _summary.txt	full log; results and config; headline numbers
R4_tolerance.csv	one row per noise case: σ , sensitivities, three tolerances
R4_sweep.csv	72 rows (6 noise cases $\times$ 12 ϵ_γ): $\delta\beta/\beta$, $\delta R/R$, d_{sys} , d_{tot} , class
figures/R4_fig1_gamma_tolerance.pdf/.png	Figure 1
Superseded	R4_20261001.183822 (pre-amendment)